Friction: Test the Surfaces · Tier Pack · Independent

Learning goal: I can compare friction on different run-out surfaces using repeated measurements and a controlled setup.

Independent Route (Book First)

Record two trials for Surface A and Surface B (practical or supplied data: A = 46, 48 cm; B = 24, 26 cm). Write a conclusion naming the numbers and one control.

Help: Surface __ stopped the block in __ cm and __ cm, compared with __ cm and __ cm on __. This shows greater friction on __ because __.

Optional Make It · Independent

Make a bar-strip comparison card

Draw two scaled paper strips (e.g. 25 mm and 47 mm) on card to represent the stopping distances on Surface B and Surface A. Label the friction conclusion in your book.

Word Help

run-out surface: the flat test surface at the bottom of the ramp where the block slows and stops

stopping distance: how far the block travels along the run-out surface before stopping, in cm

control variable: a factor kept the same so the surface comparison is fair

repeat trial: running the same surface test more than once to check consistency

Exit Check

Which stopping distance shows more friction: 24 cm or 46 cm?

Name two things we kept the same to make the surface test fair.